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First Arab Pangenome Published

Study reveals unique genomic variants for underrepresented populations

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A drive to map Arab genomic diversity has yielded the first draft of a UAE-based Arab Pangenome Reference (UPR), highlighting critical gaps in existing human genome references and offering a tailored tool for precision medicine in Arab populations. The study, published in *Nature Communications*, analyzed the genomes of 53 healthy individuals from eight Arab countries residing in the United Arab Emirates.

Using high-fidelity sequencing, the research team created 106 high-quality genome maps that represent both sets of chromosomes for each individual. These maps were more complete and accurate than existing global references. In the process, they uncovered nearly 112 million DNA bases from regions that had never been documented before, along with over 235,000 structural changes in the genome that were missing from the most commonly used genetic databases.





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Among the novel findings were 883 uniquely duplicated genes not seen in other pangenomes. Notably, the TATA-binding protein gene *TAF11L5* was duplicated in all UPR samples but absent in global references, suggesting population-specific functional relevance. Additionally, 15% of these duplicated genes are associated with recessive disorders, a key consideration in a population with elevated consanguinity rates.

The UPR also enhances variant detection sensitivity. On average, each genome contained 8.94 million population-specific small variants and up to 17,444 unique structural variants. Complex structural variations – particularly concentrated on acrocentric chromosomes – were identified in 659 genome regions.

In parallel, the mitochondrial Arab pangenome (mtUPR) revealed 1,436 base pairs of novel sequences and 623 unique variants not present in other references. These additions expand the diagnostic toolkit for mitochondrial diseases within Arab cohorts.

Comparative benchmarking demonstrated that mapping accuracy and variant recall were consistently higher with the UPR than with existing references, including CHM13 and the Human Pangenome Reference Consortium. This positions the UPR as a foundational resource for future genetic research, diagnostics, and personalized healthcare in the Middle East and beyond.